Twisty

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ABSTRACT

In the field of evolutionary robotics, the goal is to simulate evolutionary processes to design robots that are specialized for efficient task performance in real-world environments.

Evolutionary robotics does not only focus on intelligent brains but also on intelligent bodies that are well adapted to the environment in which the robot evolved. For this reason, the expansion of the morphological solution space is a fundamental part of this domain of research.

This study investigates an expansion of the morphological solution space through the introduction of twisted modules that allow diagonal relative rotations of 45°. Traditionally, modular robots evolved in simulation have relied on strictly orthogonal (90°) joint configurations to maintain mechanical simplicity and physical realizability. However, in biological systems, strictly orthogonal joint and limb orientations do not occur. Motivated by this observation, the introduction of twistiness is explored as a potential step toward increasing the realism of modular robot morphologies and examining its impact on locomotion performance.

This paper evaluates the impact of adding twisted modules to modular robots on locomotion performance. One preliminary experiment was conducted on flat terrain with brain evolution on manually adjusted morphologies, and four experiments were conducted with morphological evolution of twisted, non-twisted, and mixed populations to testing forward and backward movement on flat and rugged terrain.

The results show that introducing twistiness can lead to measurable performance improvements, particularly for forward locomotion, while backward locomotion benefits more from non-twisted configurations. This indicates that the impact of twistiness is positive in certain cases, but remains strongly dependent on the task and environment.

1 INTRODUCTION

Evolutionary robotics (ER) represents a paradigm shift from traditional monolithic designs, utilizing semi-autonomous units that can alter their physical connectivity to adapt to diverse tasks and environments [2, 7]. A foundational principle in this field is that of morphological computation, which posits that the geometric and mechanical properties of a robot's body are active participants in its control and locomotion efficiency. A critical, yet often constrained, variable in these systems is the relative orientation of module rotational axes, which is frequently restricted to simple orthogonal (90°) angles. [8].

This research investigates the impact of "twistiness", the introduction of non-standard relative rotations between modules, on locomotion performance. Unlike many studies in ER that utilize simultaneous "body-brain" co-evolution [7, 8], we decouple these processes to isolate the effects of morphology and control. We present two distinct experiments: the first evolves robot body plans using fixed Central Pattern Generators (CPGs) for control, while the second compares hand-crafted body morphologies (exclusively 90-degree rotations) against their "twisty" counterparts by evolving controllers via fully connected neural networks .

To guide this investigation, we address the following research questions:

(1) Does the introduction of twistiness improve the performance of morphological evolution?
(2) Do hand-crafted twisted robot designs show improved locomotive skills compared to non-twisted ones under controller-only evolution?

2 RELATED WORK

Our work is situated at the intersection of morphological evolution and the optimization of modular locomotion controllers. By examining morphology and control as distinct but interacting factors, we aim to isolate their individual effects on locomotion performance.

2.1 Decoupling Morphology and Control in Evolutionary Robotics

In the field of evolutionary robotics (ER), the joint optimization of morphology and control is often viewed as a pathway toward "embodied intelligence" [2]. However, simultaneous co-evolution frequently encounters a "body-brain mismatch," where morphological mutations render existing controllers obsolete, leading to a rugged fitness landscape [8]. To mitigate these complexities we propose a decoupled approach where body designs and controllers are optimized in separate loops. Our research adopts this strategy of separation to specifically isolate "twistiness" as a controlled morphological trait.

2.2 Control Frameworks for Modular Locomotion

Control strategies for modular systems range from bio-inspired rhythmic generators to data-driven neural networks. Central Pattern Generators (CPGs) are are among the most versatile solutions for their ability to produce smooth, oscillatory gaits with minimal parameters [3]. Conversely, neuroevolutionary approaches such as NeuroEvolution of Augmented Topologies (NEAT) allow for the autonomous discovery of complex, non-linear control policies by evolving the network topology alongside its weights [12]. Newer approaches for robot controls include Hypercube-Based Encoding for Evolving Large-Scale Neural Networks (HyperNEAT) [5, 11] which utilizes Compositional Pattern Producing Network (CPPN) as a genotype to create networks with mixture of activations functions and complex architectures. In order to keep complexity low

we employed CPGs in our body-evolution trials and fully connected network in variant trails. By doing so we can systematically observe how different control paradigms exploit the unique kinematics offered by twisted structures.

2.3 Performance metrics in Evolutionary Locomotion

Despite the theoretically unbounded space of possible fitness metrics, we employ a straightforward formulation to limit confounding factors. Our study utilizes traveling distance as the sole performance metric to evaluate the effectiveness of "twistiness" in overcoming environmental constraints and maximizing locomotion velocity.

3 ALGORITHM DESCRIPTION

The entire project was implemented using the ARIEL (Automated Robot Intelligence through Evolution and Learning) framework [4]. This section describes specific components of the evolution process for morphological experiments carried out in this paper. The summary of evoluition components and their corresponding hyperparamenters can be found in table 1.

Encoding. For encoding, we used a Neural Developmental Encoding (NDE). This approach utilizes neural networks to recreate robot graphs from three vectors responsible for type, connection and rotation of specific modules.

Initialization. The initial population consisted of 100 individuals created by generating three random vectors as required for NDE encoding. Additionally specific experiments required the initial population to have only modules in 0 or 90 degree rotation, in which case the rotation vector of each individual was adjusted accordingly.

Parent selection. In each evolutionary step parents were selected using exponential ranking. In this approach individuals are first sorted and assigned a rank, with higher rank corresponding to better fitness. Afterwards a probability of selection is calculated using following formula:

$$p_{selection} = \frac{1 - e^{-r}}{N} \tag{1}$$

where r is the rank of the given individual and N is the size of the population. From this ranking two individuals were chosen for crossover/mutation operations.

Crossover. After selecting the parents, blend crossover was used to create the offspring. This method does not take the gene value directly from the parents, instead it "blends" their genes to create new one. For a particular application, the crossover first chose the minimal and maximal values from both parents' corresponding alleles, calculated the difference between them and finally obtained the lower and upper bound. The α parameter was introduced to control how much the offspring can differ from it's parents.

$$LB = min - |\alpha * diff| \tag{2}$$

$$UB = max + |\alpha * diff| \tag{3}$$

$$U(LB, UB) \tag{4}$$

The probability of crossover was set to 0.7. If the crossover was not carried out, a single selected parent had the chance to mutate with probability of 0.1.

Mutation. The creep mutation mechanism used for the genetic vectors. This approach works by adding uniformly distributed noise to the entries in each NDE vector. It was applied with probability of 0.1 on the offspring of the crossover or on the newly selected parent if the crossover was not carried out. The first scenario updated the genetic code of the offspring, while the second created a new individual.

Survivor selection. Each evolution step created $3 * N$ new offspring which was combined with the original population. From this total set N best individuals were chosen to create new generation.

4 EXPERIMENTAL SETUP

This study investigates the interaction between robot morphology and controller optimization by evaluating locomotion performance across multiple morphological configurations. Two complementary experimental settings are considered: morphology evolution under fixed controllers, and controller-only optimization with fixed morphologies

Simulation Environment. All experiments were conducted in MuJoCo physics simulation [13] with the following environmental parameters:

- **Terrain**: flat or amphitheater
- **Initial position**: $[0, 0, 0.1]$, with bounding box correction applied
- **target position**: $[0, 5, 0.5]$ or $[0, -10, 0.5]$
- **Simulation duration**:
  - 30 seconds per fitness evaluation
  - 30 seconds for final visualization videos
- **Tracking**: Core body position tracked via MuJoCo geometry objects

The SimpleFlatWorld environment is a 1 by 1 flat terrain that is meant to simulate the modular robots when there are no obstacles or borders to block the path of a robot. In the experiments this environment is viewed as the base case in which there are the least possible environmental factors influencing performance.

The Amphitheater environment is a 10 by 10 rugged amphitheater shaped environment meant to simulate the modular robots under harsh environmental conditions with many obstacles on the way to the target position. In the experiments it is expected for the robots to perform worse in this environment, but it is also expected for the differences in performance between populations to be maintained.

Data Analysis and Outputs. For each morphology, the following data were collected and analyzed:

- **Best fitness**: Best fitness per generation over 20 generations
- **Diversity Score**: A value indicating how diverse a population is based on individual diversity values
- **Multi-run aggregation**: Mean fitness and diversity scores across the 30 runs

Evolution component	Method	Corresponding hyperparameter (if applicable)
Genetic encoding	NDE	-
Parent selection	Exponential ranking	-
Crossover	Blend crossover	$p_{crossover} = 0.7$; $\alpha = 0.5$
Mutation	Float creep	$p_{mutation} = 0.1$; $U(-0.1, 0.1)$
Survivor selection	Top N individuals	$N = 100$ (population size)

Table 1: Evolution components

- **Performance metrics**:
  - average best fitness for each generation
  - average diversity scores for each generation

The experimental outputs include:

- Population data files
- Fitness plots showing the best finesses for each generation
- Diversity score plots of the populations
- Video recordings of the best-performing individuals

4.1 Morphological Evolution

Morphological evolution experiments were conducted to assess the impact of body twisting on locomotion performance under evolutionary pressure. Three population compositions were evaluated:

(1) Fully non-twisted population
(2) Fully twisted population
(3) Mixed population

The fully non-twisted population consists of individuals with hinges and bricks that always have 90 degrees rotations and do not have any twisted 45 degrees rotations.

The fully twisted population consists of individuals that all have the possibility for having hinges and bricks with 45 degrees rotations. So, even if an individual in the twisted population has no hinges or bricks with 45 degrees rotations it will still be considered a twisty individual since the genotype of the individual allows for it.

The mixed population comprises both types of individuals. During evolution, non-twisty individuals cannot produce twisted offspring through mutation. However, non-twisty and twisty individuals are allowed to reproduce during evolution. If one parent is non-twisty and the other is twisty, the offspring will always be a twisty individual.

All Morphological Evolution experiments employed a Sensory-feedback-CPG-based neural controller[14] and were run for 20 generations with a population size of 100 individuals. Each experimental condition was repeated 30 times using different random seeds (41-70) to ensure statistical robustness.

The Morphological Evolution experiments were performed in two sets. One set of experiments was conducted for forward, and another for backward movement. In both sets the experiments were run in two different environments, the flat world environment and an amphitheater environment.

4.1.1 Fitness function. The fitness function returns a larger fitness, the smaller the distance is to the target position:

$$d_{\text{target}} = P_{\text{end}} - P_{\text{target}} \tag{5}$$

$$fitness = 1/(1 + d_{\text{target}}) \tag{6}$$

where the P_{end} is the final position of the modular robot at the end of the 30 seconds simulation and the P_{target} is the target position set at $[0, 5, 0.5]$ for backward locomotion and $[0, -10, 0.5]$ for forward locomotion.

4.1.2 Statistical test. Statistical comparisons between the different populations were employed using the Mann-Whitney U test [9] on the best fitnesses over 30 repetitions for each of the 20 generations, with significance level $\alpha = 0.05$ represented as a single star, $\alpha = 0.01$ as two stars and $\alpha = 0.001$ as three stars as can be seen in Figure 3.

4.1.3 Diversity Metric. For each of the 20 generations all of robots in the populations were saved as a networkx directed graphs representing the robot phenotype. With the graphs for all individuals the morphological measures were calculated using the 6d descriptor provided by a related research group ('Good genes') based on the measures from [1]. The following calculations were done with the morphological measures produced by the descriptor.

$$\bar{D} = \frac{1}{100} \sum_{i=1}^{100} \sqrt{\sum_{j=1}^{6} \left(x_{ij} - \bar{x}_j\right)^2} \tag{7}$$

Where $\bar{D}$ is the diversity score of the population, x_{ij} is the measure j for individual i and $\bar{x}_j$ is the population average of the measure j.

4.1.4 Constrains. For the morphological evolution experiments limitations were set to the allowed morphologies. The main limitation that was implemented was to disallow bodies with modules above or below the core, bricks and hinges. This limitation was set to have the evolution produce morphologies that can also be built in the real world and stop the robots from exploiting the environment to achieve higher fitness values by building tower like bodies or place modules into the ground of the terrain.

4.2 Controller-Only Evolution (Brain Optimization)

As a side study, morphological structures were fixed and only the neural controllers were optimized. This setting isolates the effect of body morphology on locomotion performance by keeping the optimization procedure and controller architecture identical across all variants. The morphologies were hand-crafted variants of a common morphological template, hereafter referred to as *gecko*.

4.2.1 Evaluated Morphologies. Six distinct gecko morphologies were evaluated, comprising five twisted configurations and one non-twisted baseline:

(1) **gecko**: Heuristic-based twisted configuration, serving as the current standard morphology in ARIEL [4]
(2) **gecko_untwisted**: Original non-twisted body morphology
(3) **gecko_good**: Optimized twisted morphology variant
(4) **gecko_doubletwist**: Double-twisted body configuration
(5) **gecko_doubletwist_turtle**: Turtle-inspired double-twisted variant
(6) **gecko_front**: Front-biased twisted configuration

All morphologies were implemented using the ARIEL framework [4], which provides a modular system for defining robot body phenotypes and conducting evolutionary robotics experiments. Each morphology was evaluated independently to determine how structural differences influence locomotion capability.

To support reproducibility and to enable a clear qualitative comparison between the fixed body designs, we developed a dedicated visualization tool that automatically extracts structural information directly from the MuJoCo model specification. For each morphology, the tool instantiates the body in a standardized simulation environment and generates three-dimensional schematic renderings in which individual modules are represented as rigid blocks connected according to the underlying kinematic tree.

In addition to spatial layout, the visualization explicitly encodes local module orientation by rendering body-fixed coordinate axes for each articulated component. This makes relative rotations between modules (i.e., twistiness) directly observable and allows structural differences between morphologies to be inspected independently of controller behavior.

Figure 1 shows the resulting schematic visualizations for all evaluated gecko morphologies.

4.2.2 Neural Controller Architecture. Each gecko morphology was controlled by a feedforward neural network that mapped joint positions to motor commands. The controller architecture consisted of:

- **Input**: Joint position data, with dimensionality determined by the `qpos` vector
- **Hidden layers**: Three fully connected layers with 32 neurons each
- **Activation function**: Exponential Linear Units (ELU)
- **Output**: Motor control signals for all actuators

The Neural network setup is further visualized in Fig. 2. Keep in mind that Input and output sizes are morphology-dependent.

4.2.3 Optimization Strategy. Controller parameters (neural network weights and biases) were optimized using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [6], implemented through the Nevergrad library [10]. The optimization parameters were identical across all morphologies:

- **Evaluation budget**: 26,000 fitness evaluations per experiment
- **Objective**: Maximize forward displacement along the Y-axis
- **Fitness function**: Distance traveled from the starting position (negative Y-displacement)

4.2.4 Experimental Protocol. For each morphology, three independent optimization runs were conducted using different random seeds (42, 43, and 44). Neural network weights were randomly initialized at the start of each run.

Fitness function. The fitness function maximized forward locomotion distance along the Y-axis:

$$f(\theta) = y_{\text{start}} - y_{\text{end}} \tag{8}$$

where y_{start} and y_{end} represent the robot's Y-coordinate at the beginning and end of a 15-second simulation episode, respectively, and θ denotes the neural network parameters. Negative Y-displacement corresponds to forward motion in the simulation coordinate system.

5 EXPERIMENTAL RESULTS

5.1 Gecko Brain Experiment

The results of the gecko experiments showed that the doubletwist turtle gecko (visualized in Fig. 1) had a significantly higher fitness compared to the other gecko morphologies. The fitness plots for all the different gecko morphologies can be seen in Figure 7 and Figure 8. The plots show that the average fitness of the doubletwist turtle gecko is highest on the flat terrain over the 250k iterations compared to the other morphologies. The plots also show that the original gecko had the second highest fitness and the untwisted gecko had a similar fitness curve as the doubletwist gecko.

5.2 Morphological Evolution

The results from the flat world experiments were visualized in Figure 3 and Figure 5.

The results for forward movement are found in Figure 5, Figure 5c and Figure 5a show that the twisted population had a significantly higher fitness compared to the non-twisty and the mixed populations over all generations. Meanwhile, between the mixed population and the non-twisty population only a small significant difference could be found for the last four generations as seen in Figure 5b. Notably, the confidence interval for the non-twisty population shows a higher stability than both the twisted and the mixed populations.

The diversity plots in Figure 11 show that for the first few generations the twisty and mixed populations have a significantly higher diversity score, while the non-twisty population shows to have a higher diversity score over the final generations. Notably, the non-twisty population shows a large confidence interval for the last eight generations indicating differences in diversity scores between repetitions.

The results for the backward movement are found in Figure 3. Figure 3c and Figure 3b show that for backward movement the non-twisty population performed significantly better than the twisted and mixed populations over all generations. The mixed population was separately found to be significantly better than the twisted population as well, as shown in Figure 3a. It can also be seen that the confidence interval is stable for all populations over all generations.

The diversity plots can be seen in Figure 9. The plots show that The twisty population had a significantly higher diversity score than the non-twisty population over all generations and there was no significant difference between twisty and mixed populations. The plots show a similar confidence interval for all of the populations indicating the evolution is consistent over repetitions.

The second set of experiments that were conducted on the amphitheater environment. The results from the experiments can be seen in Figure 6 and Figure 4.

The results for the forward movement are found in Figure 6. None of the populations had any significant differences. Only with the confidence intervals small differences can be seen between the populations, but all of the intervals show stability over all generations.

The diversity plots in Figure 12 show that there were also no significant differences among the different populations. But, while the confidence interval for the mixed and twisted populations are more stable it can be seen in Figure 12c that the confidence interval is highly unstable for the non-twisty population.

Finally, the plots for the backward movement in the amphitheater are found in Figure 4. The non-twisty population was found to have a significantly higher fitness than the twisted population as seen in Figure 4c. No significant difference was found between the non-twisty and mixed populations, which is seen in Figure 4b and the mixed population had a significantly higher fitness than the twisted population only for the first ten generations shown in Figure 4a. The confidence intervals for all of the populations are similar over all generations.

The diversity plots are found in Figure 10. It can be seen that the diversity score of the non-twisty and the mixed populations were significantly higher than the diversity score of the twisty population for the first five generations as can be seen in Figure 10c and Figure 10b. No significant differences were found between the non-twisty and mixed populations as is shown in Figure 10a. The confidence interval was stable for the mixed and non-twisty populations, but for the twisty population during the last few generations the interval can be seen to be more unstable.

6 ANALYSIS AND DISCUSSION

The study in this paper was conducted to answer the research question of whether adding the possibility for having twisted hinges and bricks to modular robot would result in a higher performance through evolution. The initial hypothesis in this paper was that twisted morphologies will perform better than non-twisty individuals both in a flat terrain and a rugged terrain.

The preliminary results with prebuilt gecko morphologies showed that on a flat terrain some of the morphologies with twisted bricks or hinges were able to learn faster on the flat terrain and move a larger distance than the non-twisty gecko. However, these results were only based on three repetitions per morphology. Therefore, these results are not conclusive and are presented as results from initial testing.

The results from the morphological evolution on flat terrain show that differences in performance are task dependent. For forward locomotion, the twisty population had a significantly higher fitness and low diversity scores. Both the diversity scores and fitness also have small confidence intervals for the twisty population, indicating that the twisty population converges. On the other hand, for the non-twisty populations, the performance has a small confidence interval, while the diversity scores have a large confidence interval. This indicates that even though the non-twisty populations have different diversity scores, the differences in fitness between the populations remain small.

for backward locomotion the non-twisty perform significantly better, while the twisty populations have a significantly higher diversity score. Furthermore, for all populations the performance has a small confidence interval, showing that the results are consistent. The diversity scores show that the non-twisty populations also have a small confidence interval for the last generation with a small diversity score, showing that the non-twisty populations have convergence and have a consistently better performance that the twisted populations.

Meanwhile, the results of the morphological evolution on the rugged amphitheater terrain showed that differences between populations becomes smaller on rugged terrain compared to a flat terrain. For forward locomotion there are no significant differences between the fitnesses and the diversity scores of the populations. But the non-twisty populations do show large instability with the diversity scores, while the performance of the non-twisty populations are stable. Indicating that for the non-twisty populations, large differences in diversity does not yield large differences in performance.

For backward locomotion, the non-twisty population had a significantly higher fitness than the twisted population, but there was no significant difference with the mixed population. The confidence intervals show that the results are consistent over the repetitions. From the diversity plots it can be seen that the twisty populations are instable near the last few generations and have a lower diversity score, while the non-twisty populations have a more stable diversity score with a higher diversity score. Meaning that the non-twisty populations has consistently higher diversity and a better performance than twisty populations with backward locomotion on the amphitheater terrain.

It is important to add that it is possible the results were impacted by the choice of environment and the limitations. The main limitations were to disallow bodies with modules above or below the core, bricks and hinges. These limitations were set to make the robots similar to the real life versions that can be build and to avoid making bodies that can produce behavior classified as "cheating", for instance, robots with bodies that initialize high above or below the surface of the environment to move closer to the target position. On the other hand, the experiments were only conducted on two environments, of which one was chosen for being flat and another for being rugged. Therefore, the findings should be interpreted in the context of the chosen environments and constraints, as different settings may yield different results.

For future studies, focus can be placed on implementing twisty robots in the real world and the results found from this study can be used to determine the impact of the reality gap. So that the impact of twisted modules can also be studied from a real world perspective. The eventual goal is to base the results found through this study and future studies on defining a baseline for twisty morphologies and determining the tasks in which twisted morphologies will have the most impact.

7 CONCLUSION

In conclusion, this study meant to explore the potential of twisted modules in the field of evolutionary robotics by answering the research questions of whether twistiness improves robot locomotion and whether adding twistiness can improve non-twisty morphologies.

On flat terrain, twisted modules achieved higher performance for forward locomotion, whereas non-twisted modules performed better for backward locomotion. In the amphitheater environment, twisted and non-twisted designs performed equally for forward movement, but non-twisted bodies outperformed twisted ones when moving backwards. On the other hand, in the gecko-based experiments on flat terrain, adding twistiness to the body design led to improved locomotion performance compared to the non-twisted baseline.

These findings demonstrate that twistiness provides benefits in specific cases, such as forward motion on flat terrain and for certain hand-crafted morphologies, but it is not universally advantageous. Adding extra degrees of freedom can actively hamper performance when the resulting morphological complexity is misaligned with the task or environment. As such, increased morphological expressiveness should be viewed as a trade-off rather than an inherent improvement, requiring careful justification rather than default inclusion.

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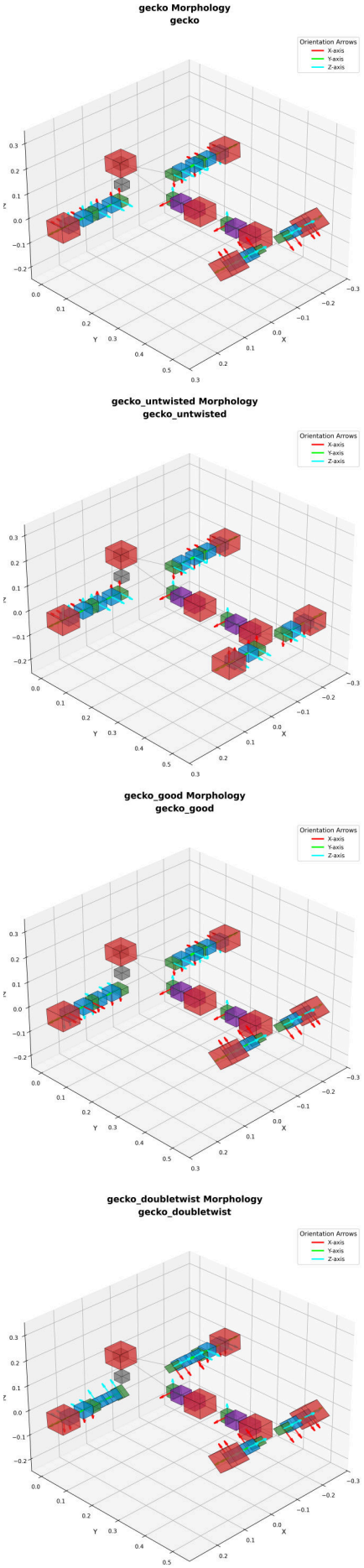
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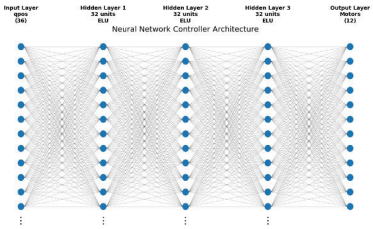
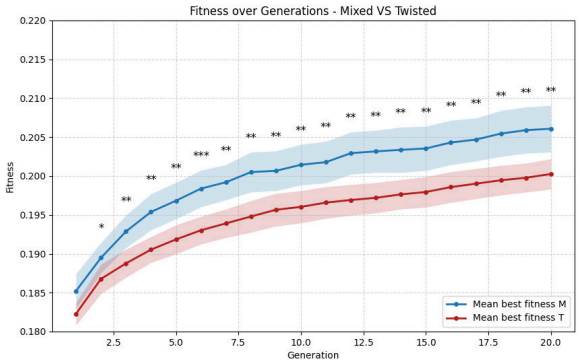
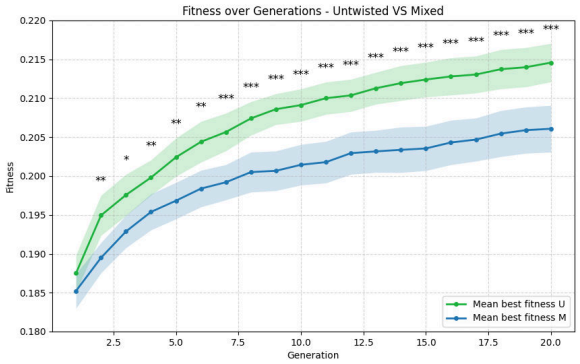


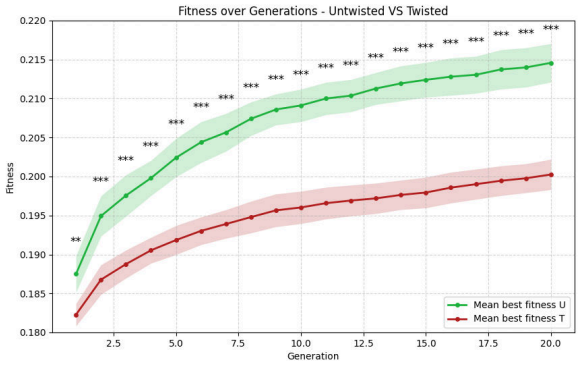
Figure 2: Visualisation of Controller NN



(a) Plot of the mixed (blue) and twisted (red) populations on the flat world environment for backward movement.

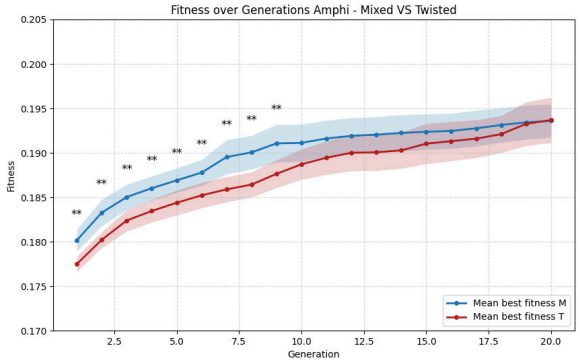


(b) Plot of the Untwisted (green) and mixed (blue) populations on the flat world environment for backward movement.

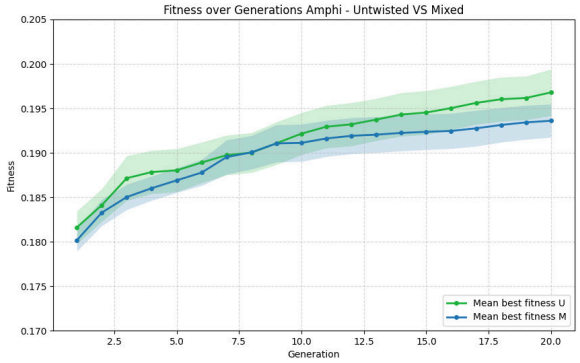


(c) Plot of the Untwisted (green) and twisted (red) populations on the flat world environment for backward movement.

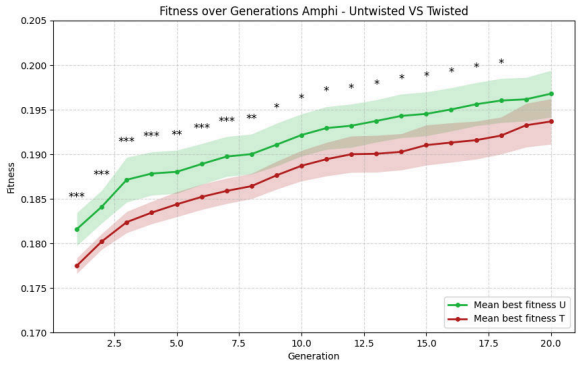
Figure 3: fitness plots for morphological evolution in flat world environment for backward movement.



(a) Plot of the mixed (blue) and twisted (red) populations on the amphitheater environment for backward movement.

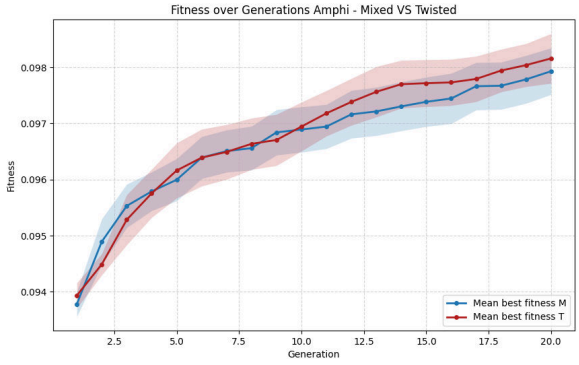


(b) Plot of the mixed (blue) and untwisted (green) populations on the amphitheater environment backward movement.

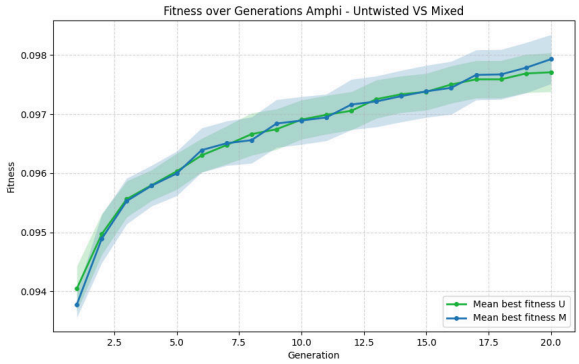


(c) Plot of the twisted (red) and untwisted (green) populations on the amphitheater environment backward movement.

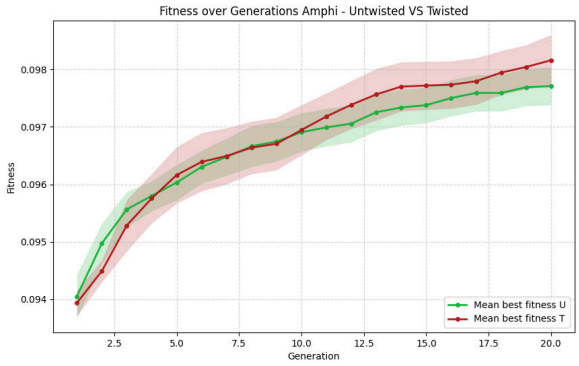
Figure 4: fitness plots for morphological evolution in amphitheater world environment for backward movement.



(a) Plot of the mixed (blue) and twisted (red) populations on the amphitheater environment for forward movement.

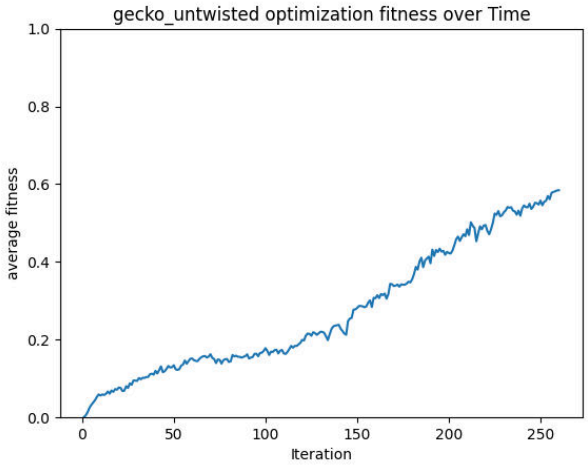


(b) Plot of the mixed (blue) and untwisted (green) populations on the amphitheater environment forward movement.

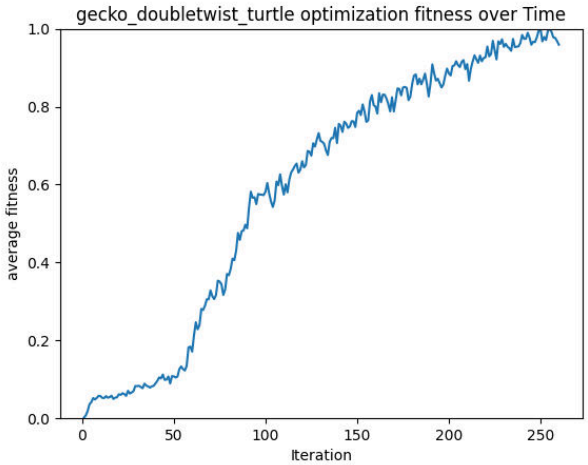


(c) Plot of the twisted (red) and untwisted (green) populations on the amphitheater environment forward movement.

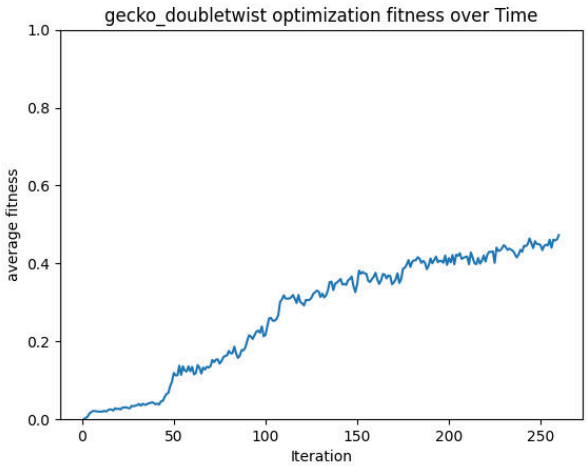
Figure 6: fitness plots for morphological evolution in amphitheater world environment for forward movement.



(a) Fitness plot of the untwisted gecko brain evolution over 250k iterations.

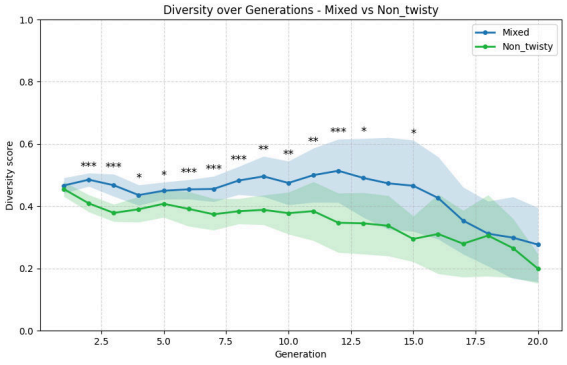


(b) Fitness plot of the doubletwist turtle gecko brain evolution over 250k iterations.

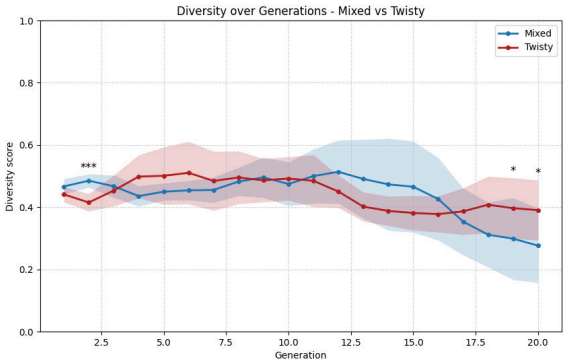


(c) Fitness plot of the gecko doubletwist brain evolution over 250k iterations.

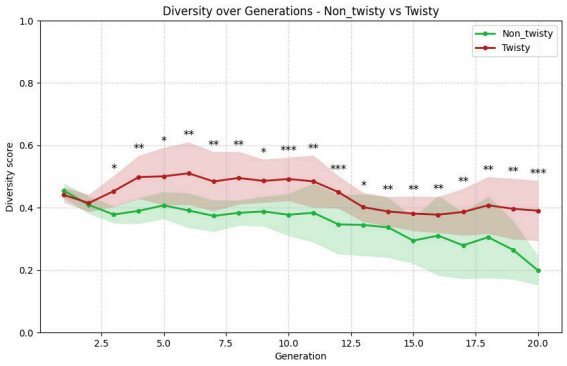
Figure 7: fitness plots for the untwisted gecko, doubletwist gecko and doubletwist turtle gecko morphologies with neural network brains.



(a) Diversity scores of Mixed and non-twisty populations for the flat terrain experiment with backward movement.

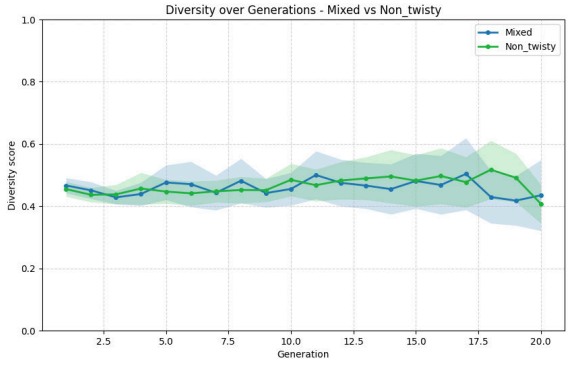


(b) Diversity scores of Mixed and Twisty populations for the flat terrain experiment with backward movement.

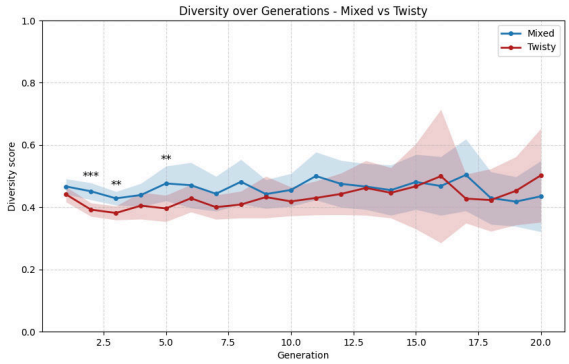


(c) Diversity scores of non-twisty and Twisty populations for the flat terrain experiment with backward movement.

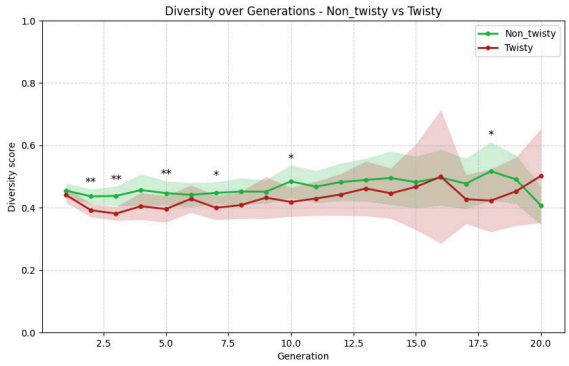
Figure 9: diversity score plots for backward movement in the flat terrain experiment.



(a) Diversity scores of mixed and non-twisty populations for the amphitheater terrain experiment with backward movement.

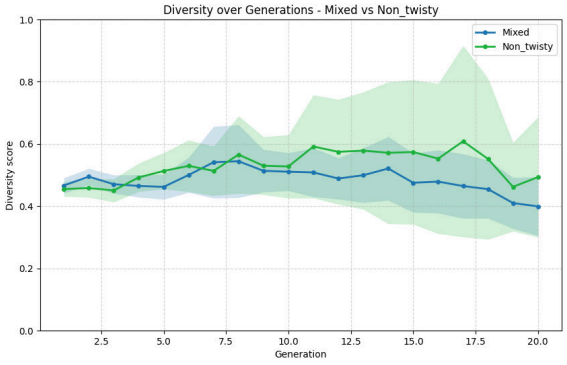


(b) Diversity scores of Mixed and Twisty populations for the amphitheater terrain experiment with backward movement.

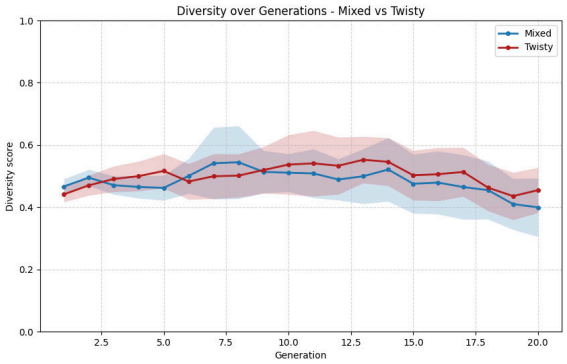


(c) Diversity scores of non-twisty and Twisty populations for the amphitheater terrain experiment with backward movement.

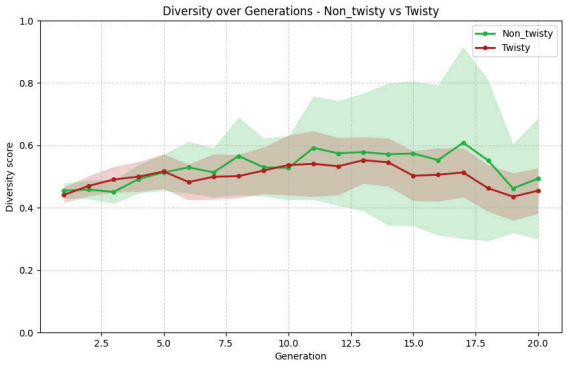
Figure 10: diversity score plots for backward movement in the Amphitheater terrain experiment.



(a) Diversity scores of mixed and non-twisty populations for the amphitheater terrain experiment with forward movement.

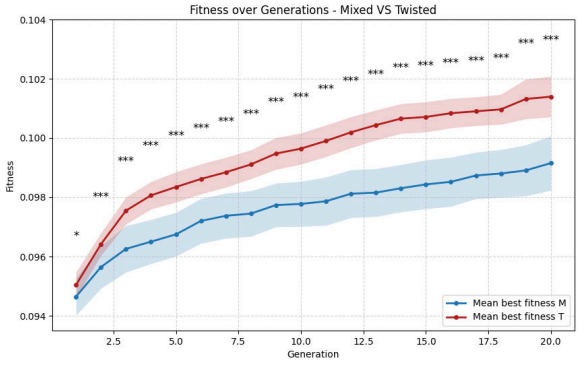


(b) Diversity scores of Mixed and Twisty populations for the amphitheater terrain experiment with forward movement.

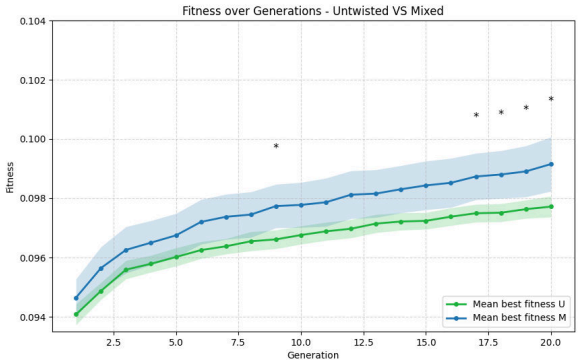


(c) Diversity scores of non-twisty and Twisty populations for the amphitheater terrain experiment with forward movement.

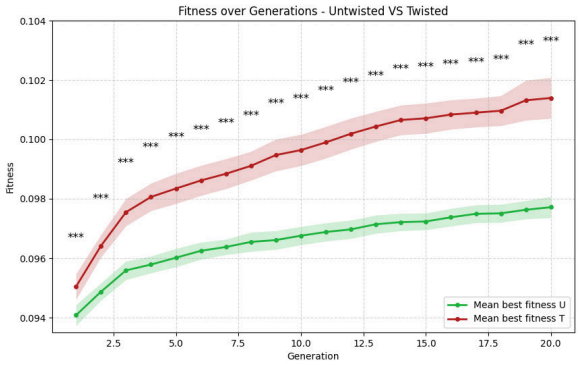
Figure 12: diversity score plots for forward movement in the Amphitheater terrain experiment.



(a) Plot of the mixed (blue) and twisted (red) populations on the flat world environment for forward movement.



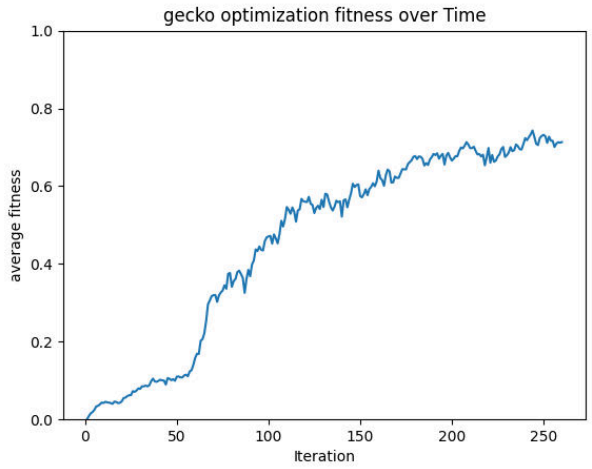
(b) Plot of the mixed (blue) and untwisted (green) populations on the flat world environment for forward movement.



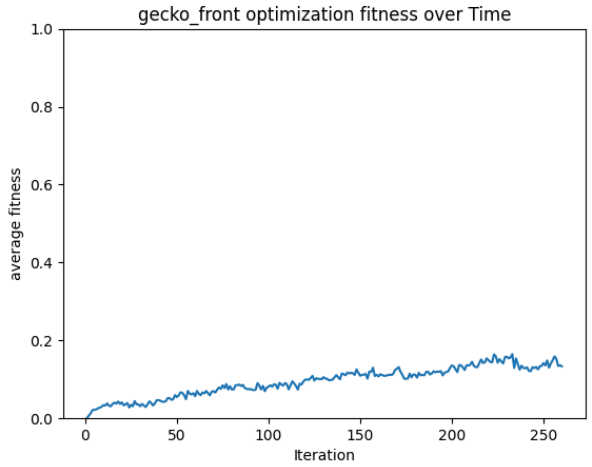
(c) Plot of the twisted (red) and untwisted (green) populations on the flat world environment for forward movement.

Figure 5: fitness plots for forward movement in the flat terrain experiment.

8 APPENDIX



(a) Fitness plot of the gecko brain evolution over 250k iterations.

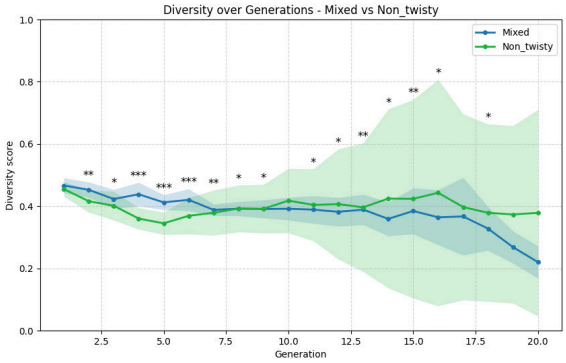


(b) Fitness plot of the gecko front brain evolution over 250k iterations.

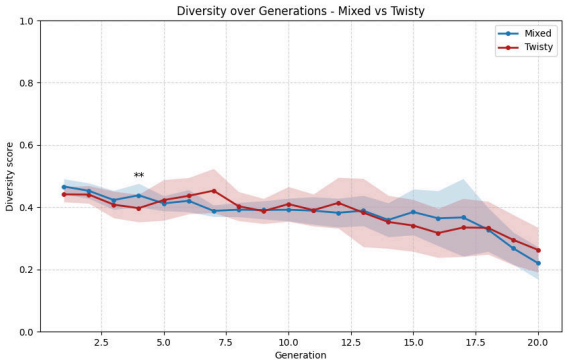


(c) Fitness plot of the good gecko brain evolution over 250k iterations.

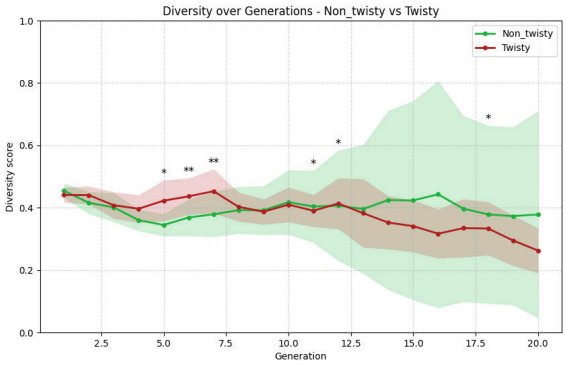
Figure 8: fitness plots for the gecko, front gecko and good gecko morphologies with neural network brains.



(a) Diversity scores of mixed and non-twisty populations for the flat terrain experiment with forward movement.



(b) Diversity scores of Mixed and Twisty populations for the flat terrain experiment with forward movement.



(c) Diversity scores of non-twisty and Twisty populations for the flat terrain experiment with forward movement.

Figure 11: diversity score plots for forward movement in the flat terrain experiment.